Complex Engineering Problem Evaluation Checklist (P1–P7)

Tick ✓ "Yes" if the problem demonstrates the characteristic.
A problem may satisfy **P1 + at least two** of P2–P7 to qualify as a CEP (BAETE/Washington Accord).

P1 – Depth of Knowledge Required

Does the problem require in-depth engineering knowledge from K3, K4, K5, K6, or K8?

- ☐ Requires engineering fundamentals beyond textbook examples
- ☐ Requires specialist knowledge (advanced course topics)
- ☐ Requires design knowledge, interpretation, or innovation
- ☐ Requires use of engineering practice tools/technologies
- ☐ Requires consulting research papers or advanced references

If mostly routine or solvable with K1–K2 only → NOT P1.

P2 – Range of Conflicting Requirements

Does the problem involve multiple constraints that conflict with each other?

- ☐ Trade-offs between performance, cost, power, size, speed, or reliability
- ☐ Multiple design objectives must be optimized simultaneously
- ☐ Conflicting user/technical requirements
- ☐ Need to compromise or prioritize among constraints

If only one constraint → NOT P2.

P3 – Depth of Analysis Required

Does the problem require abstract thinking or creation of a new/advanced model?

- ☐ No obvious or ready-made solution exists
- ☐ Requires developing a model, simulation, or mathematical abstraction
- ☐ Requires original interpretation of data or system behavior
- ☐ Needs creative problem decomposition or analytical techniques

If solvable directly by formulas or standard methods → NOT P3.

P4 – Familiarity of Issues

Does it involve issues that are rare, unfamiliar, or not routinely encountered by students?

- ☐ New or emerging technology
- ☐ Uncommon application or scenario

☐ Complex behavior not typically covered in class examples

☐ Unpredictable or unusual conditions

If it resembles typical homework/class examples → NOT P4.

P5 – Extent of Applicable Codes

Can the solution be fully obtained by simply following standards, rules, or guidelines?

☐ Existing standards (IEEE/IEC/ITU/ISO) do NOT fully cover the problem

☐ Requires interpreting or adapting standards

☐ Requires integrating multiple partial standards

☐ Problem needs custom rules or creative extensions beyond codes

If standards fully dictate the solution → NOT P5.

P6 – Stakeholder Diversity & Conflicts

Are there multiple stakeholders with differing needs/constraints?

☐ Users, engineers, management, regulators, and/or society involved

☐ Stakeholder requirements conflict with each other

☐ Requires negotiation/compromise among stakeholders

If only one stakeholder (e.g., the student) → NOT P6.

P7 – Interdependence of Systems

Is the problem composed of many interdependent subsystems?

☐ Multiple subsystems must work together (hardware, software, control, power, etc.)

☐ Failure in one subsystem affects others

☐ Requires system-level thinking

☐ Problem cannot be solved by focusing on only one component

If simple or single component → NOT P7.

Knowledge Profile (K1–K8) Evaluation Checklist

Tick **Yes/No** for each Knowledge Attribute based on what the CLO/activity requires.

K1 – Natural Sciences

Physics, chemistry, natural science principles applied to engineering.
Does the problem require:

- ☐ Applying physics (e.g., electromagnetics, optics, semiconductor physics)?
- ☐ Understanding material or natural science behavior?
- ☐ Using scientific principles to explain device/system behavior?
- ☐ Scientific reasoning beyond memorization?

K2 – Mathematics & Computing Fundamentals

Mathematics, numerical methods, probability, discrete math, computing basics.
Does the problem require:

- ☐ Using calculus, differential equations, linear algebra?
- ☐ Probability, statistics, random processes?
- ☐ Discrete mathematics or basic algorithms?
- ☐ Numerical analysis, simulation, or approximation methods?

K3 – Engineering Fundamentals

Core engineering theory every ECE engineer must know.
Does the problem require:

- ☐ Circuit theory, digital logic, signals, control basics?
- ☐ Applying fundamental engineering laws/theorems?
- ☐ Basic modeling/analysis from foundational courses?

K4 – Specialist / Advanced Engineering Knowledge

Deep knowledge at the forefront of the discipline.
Does the problem require:

- ☐ Advanced technologies (VLSI, DSP, renewable energy, power systems, RF)?
- ☐ Specialization-level theory or advanced tools?
- ☐ Knowledge from higher-level or elective courses?

K5 – Design Knowledge

Knowledge needed to design engineering systems.

Does the problem require:

- ☐ Engineering design decisions with constraints?
- ☐ Trade-offs among cost, performance, reliability, etc.?
- ☐ Synthesizable circuits, PCB design, system design thinking?
- ☐ Creating, optimizing, or modifying a design?

K6 – Engineering Practice (Technology)

Tools, equipment, standards, practical methods.

Does the problem require:

- ☐ Using lab tools (oscilloscope, logic analyzer, power analyzer, etc.)?
- ☐ Using software tools (MATLAB, SPICE, HDL, simulation)?
- ☐ Understanding industry practices, protocols, or implementation rules?
- ☐ Practical troubleshooting or measurement?

K7 – Engineering & Society

Ethics, safety, environmental, social, sustainability considerations.

Does the problem require:

- ☐ Ethical judgment or safety considerations?
- ☐ Environmental/sustainability thinking?
- ☐ Societal or economic impact assessment?
- ☐ Professional responsibility?

K8 – Research Literature Engagement

Using or understanding scholarly research.

Does the problem require:

- ☐ Reading IEEE/ACM/Elsevier papers?
- ☐ Comparing existing research methods or technologies?
- ☐ Using research findings to justify design decisions?
- ☐ Exploring emerging technologies not fully covered in textbooks?